A complete matrix characterization of entire Fock integral kernels

Autonomous proof-discovery packet for arXiv:1907.00574

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Abstract

Cao, Li, Shen, Wick, and Yan ask which entire two-variable kernels induce bounded Gaussian integral operators on $\mathcal{F}^2(\mathbb{C}^n)$. We give a complete answer to the question as written. In the normalized monomial expansion of the kernel, the coefficient array is exactly the operator matrix; boundedness is therefore equivalent to boundedness of that matrix on $\ell^2(\mathbb{N}_0^n)$, with equality of norms. We also give an equivalent finite coherent-state inequality involving only kernel values. An explicit block example shows that separate Fock-slice and pointwise-growth conditions are not sufficient.

1 Source problem

Let

$$d\lambda(w) = \pi^{-n} e^{-|w|^2}\, dw$$

and let $\mathcal{F}^2(\mathbb{C}^n)$ be the corresponding Gaussian Fock space. The source asks for a characterization of the entire functions $K(z, w)$ on $\mathbb{C}^{2n}$ for which

$$T_K F(z) = \int_{\mathbb{C}^n} K(z, \bar{w}) F(w)\, d\lambda(w) \tag{1}$$

is bounded on $\mathcal{F}^2(\mathbb{C}^n)$ [1]. Figure 1 reproduces the exact question and its $T(1)$ motivation.

In the theory of singular integrals in harmonic analysis, it is well-known (see [10, 27]) that the famous "$T(1)$" theorem of David and Journé gives necessary and sufficient conditions for generalized Calderon-Zygmund operators to be bounded on $L^2(\mathbb{R}^n)$. We propose the following open problem on the Fock space $\mathscr{F}^2(\mathbb{C}^n)$ (see also Proposition 2.1).

Open problem: Characterize those entire functions $K_T(z, w)$ on $\mathbb{C}^{2n}$ such that the integral operator

$$TF(z) = \int_{\mathbb{C}^n} K_T(z, \bar{w}) F(w) d\lambda(w), \qquad z \in \mathbb{C}^n$$

is bounded on $\mathscr{F}^2(\mathbb{C}^n)$.

Figure 1: The open problem on page 13 of arXiv:1907.00574.

For a multi-index $\alpha \in \mathbb{N}_0^n$, put

$$e_\alpha(z) = \frac{z^\alpha}{\sqrt{\alpha!}}.$$

The e_α form an orthonormal basis of $\mathcal{F}^2(\mathbb{C}^n)$, and every entire K has a unique normalized expansion

$$K(z,w) = \sum_{\alpha,\beta\in\mathbb{N}_0^n} a_{\alpha\beta} e_\alpha(z) e_\beta(w). \tag{2}$$

2 Full characterization

Theorem 2.1. *For an entire function K with expansion (2), the following are equivalent.*

1. *The Gaussian integral (1), in its ordinary Lebesgue interpretation, defines a bounded operator on $\mathcal{F}^2(\mathbb{C}^n)$.*
2. *The matrix $A = (a_{\alpha\beta})$ defines a bounded operator on $\ell^2(\mathbb{N}_0^n)$.*
3. *There is a constant C such that, for all finite point families $z_1, \dots, z_r, w_1, \dots, w_s \in \mathbb{C}^n$ and scalar families $\xi_1, \dots, \xi_r, \eta_1, \dots, \eta_s$,*

$$\left|\sum_{i=1}^{r}\sum_{j=1}^{s} \xi_i \overline{\eta_j} K(z_i, \overline{w_j})\right| \tag{3}$$
$$\leq C \left(\sum_{i,k=1}^{r} \xi_i \overline{\xi_k} e^{z_i \cdot \overline{z_k}}\right)^{1/2} \left(\sum_{j,l=1}^{s} \eta_j \overline{\eta_l} e^{w_j \cdot \overline{w_l}}\right)^{1/2}.$$

Moreover,

$$\|T_K\| = \|A\| = \inf\{C : (3) \text{ holds}\}.$$

Thus the answer is intrinsic in either Taylor coefficients or kernel values. No asymptotic or additional regularity hypothesis is needed.

3 Proof

The Gaussian moments satisfy

$$\int_{\mathbb{C}^n} \overline{e_\beta(w)} e_\gamma(w)\, d\lambda(w) = \delta_{\beta\gamma}. \tag{4}$$

Consequently, whenever (1) is bounded, angular integration against the monomial e_β extracts the βth coefficient in the second variable and gives

$$T_K e_\beta = \sum_\alpha a_{\alpha\beta} e_\alpha. \tag{5}$$

The Lebesgue interpretation in the theorem justifies the Gaussian moments; equivalently one may first test on polynomials. Hence, under the unitary coefficient map

$$U : \mathcal{F}^2(\mathbb{C}^n) \longrightarrow \ell^2(\mathbb{N}_0^n), \qquad U\left(\sum_\alpha f_\alpha e_\alpha\right) = (f_\alpha)_\alpha,$$

we have $UT_KU^{-1} = A$. This proves (1) $\Rightarrow$ (2) and $\|A\| = \|T_K\|$.

Conversely, assume A is bounded. Define $T = U^{-1}AU$. The coherent vector

$$v(z) = (e_\alpha(z))_\alpha$$

belongs to $\ell^2(\mathbb{N}_0^n)$ and satisfies $\|v(z)\| = e^{|z|^2/2}$. Boundedness of A and Cauchy–Schwarz imply locally uniform convergence of

$$v(z)^T A\overline{v(w)} = \sum_{\alpha,\beta} a_{\alpha\beta} e_\alpha(z)\overline{e_\beta(w)} = K(z,\bar{w}).$$

For fixed z, the anti-holomorphic slice belongs to $L^2(d\lambda)$ because A^T is bounded. Therefore the integral in (1) converges absolutely by Cauchy–Schwarz, and (4), first for polynomials and then by density, gives $T_K = T$. This proves (2) $\Rightarrow$ (1).

If A is bounded, then

$$\sum_{i,j} \xi_i \overline{\eta_j} K(z_i, \overline{w_j}) = X^T A\overline{Y}, \quad X = \sum_i \xi_i v(z_i), \quad Y = \sum_j \eta_j v(w_j).$$

Cauchy–Schwarz gives (3), since

$$\|X\|^2 = \sum_{i,k} \xi_i \overline{\xi_k} e^{z_i \cdot \overline{z_k}},$$

and similarly for Y. This proves (2) $\Rightarrow$ (3) with $C = \|A\|$.

Finally, coherent vectors have dense linear span: a vector orthogonal to all of them would define an entire Fock function vanishing everywhere, hence would be zero. Therefore (3) defines a bounded bilinear form on dense subspaces of two copies of $\ell^2(\mathbb{N}_0^n)$, and it extends with norm at most C. Its matrix on the standard basis is forced by the Taylor coefficients to be A. Thus (3) $\Rightarrow$ (2) and the least constant is $\|A\|$, completing the proof.

Corollary 3.1. *The operator T_K is compact exactly when A is compact. It is Hilbert–Schmidt exactly when*

$$\sum_{\alpha,\beta} |a_{\alpha\beta}|^2 < \infty,$$

and in that case this sum is $\|T_K\|_{\mathrm{HS}}^2$. For every bounded kernel,

$$|K(z,\bar{w})| \le \|T_K\| e^{(|z|^2+|w|^2)/2}. \tag{6}$$

Proof. The ideal statements follow from unitary equivalence. The growth estimate is the coherent-vector bound with one point in each family. □

4 Why separate kernel tests are insufficient

The source's Proposition 2.1 records necessary Fock-slice and exponential growth conditions. Even together, they do not imply boundedness.

Proposition 4.1. *There is an entire kernel K on $\mathbb{C}^2$ such that every slice belongs to $\mathcal{F}^2(\mathbb{C})$ and*

$$|K(z,w)| \le Ce^{|z|^2+|w|^2},$$

but T_K is unbounded. Its normalized coefficient matrix has every row and every column of ℓ^2-norm one.

Proof. Partition $\mathbb{N}_0$ into consecutive blocks I_k with $|I_k| = k$ and set

$$a_{ij} = \begin{cases} k^{-1/2}, & i,j \in I_k, \\ 0, & \text{otherwise.} \end{cases}$$

Since $|a_{ij}| \le 1$, the normalized double power series converges locally uniformly and defines an entire K. Every row and column has squared norm $k \cdot k^{-1} = 1$. For fixed z, the squared Fock norm of the second slice is

$$\sum_k \left| \sum_{i \in I_k} e_i(z) \right|^2 \le \sum_k k \sum_{i \in I_k} |e_i(z)|^2 \le \sum_{i \ge 0} (i+1) \frac{|z|^{2i}}{i!} < \infty.$$

The other slices are identical. Also, Cauchy–Schwarz with weights $(i+1)^{-1}$ gives

$$\sum_{i \ge 0} \frac{r^i}{\sqrt{i!}} \le C\big((r^4 + 3r^2 + 1)e^{r^2}\big)^{1/2} \le C' e^{r^2},$$

which proves the stated coarse growth bound for the double series.

On $\ell^2(I_k)$, however, the matrix is $k^{-1/2} J_k$, whose norm is $\sqrt{k}$. The full matrix is unbounded, so Theorem 2.1 shows that T_K is unbounded. □

5 Proof intuition and scope

Proof intuition. The Gaussian weight makes the normalized monomials exactly orthonormal. Integrating the kernel against one monomial therefore selects one column of its Taylor array. Nothing is lost: an arbitrary entire kernel represents an arbitrary infinite operator matrix. The coherent-state condition is the same fact expressed without coordinates; it asks that all finite matrix coefficients be controlled in the natural reproducing-kernel norm.

The theorem fully answers the displayed all-entire-kernels question. The preceding $T(1)$ discussion also suggests a more ambitious goal: local size, smoothness, cancellation, and testing conditions for a specified Fock Calderón–Zygmund class. No such class or hypotheses are specified in the source, and the class of all entire kernels realizes all of $B(\ell^2(\mathbb{N}_0^n))$. A local $T(1)$ theory is therefore a distinct future program, not an omitted step here.

Six focused routes were audited, including Bargmann conjugation, coherent states, Schur-type slice tests, and a local $T(1)$ upgrade. Bounded primary-source searches through 11 August 2026 found no later general answer to the displayed problem. The coefficient argument is elementary RKHS theory and may be folklore, so novelty confidence is low; the claim is a literal full characterization, not a claim of a new David–Journé theorem.

References

[1] G. Cao, J. Li, M. Shen, B. D. Wick, and L. Yan, *A Boundedness Criterion for Singular Integral Operators of Convolution Type on the Fock Space*, J. Geom. Anal. **31** (2021), 6533–6552; arXiv:1907.00574.

[2] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.